Recursiveness, Switching, and Fluctuations in a Replicating Catalytic Network

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(Dated:)

A protocell model consisting of mutually catalyzing molecules is studied in order to investigate how chemical compositions are transferred recursively through cell divisions under replication errors. Depending on the path rate, the numbers of molecules and species, three phases are found: fast switching state without recursive production, recursive production, and itinerancy between the above two states. The number distributions of the molecules in the recursive states are shown to be log-normal except for those species that form a core hypercycle, and are explained with the help of a heuristic argument.

PACS numbers:

arXiv:cond-mat/0302313v1 15 Feb 2003

In a cell, a huge number of chemicals is synthesized by mutual catalyzation leading to replication of molecules that allow a cell to grow until it is large enough to divide into two. How the underlying reaction networks give rise to the recursive production of cells is an important question, not only when considering the origin of life [1, 2] but also when trying to understand the general features of a modern cell's biochemical reaction dynamics.

As a simple prototype of a reproducing cell, let us consider a set of chemicals with some catalytic activities. How can such a system consisting of chemicals connected by a catalytic reaction network sustain recursive production? Are there any generic properties in the dynamics and fluctuations of such reproducing systems?

These questions were originally addressed in connection with the origin of life. Eigen and Schuster proposed the hypercycle[1] as a mechanism to overcome an inevitable loss in the catalytic activities through mutations, while Dyson[2] argued that it is possible for a collection of chemicals to be sustained by mutual catalytic activity. Although the hypercycle itself may be susceptible to destruction by parasitic molecules, i.e., molecules which are catalyzed by the hypercycle species but themselves do not catalyze other molecules, it was later shown that compartmentalization by a cell structure or localized patterns in reaction-diffusion systems may suppress the invasion of parasitic molecules [3, 4].

Here, we study a simple model of mutually catalyzing molecules and classify the biochemical states according to their ability for recursive reproduction. Besides fixed recursive states, we find fast switching states and several quasi-recursive states that allow for both recursive reproduction and evolution. Last, we study the characteristics of the number distributions of the molecular species in these replicating cells.

We envision a (proto)cell containing k molecular species with some of the species possibly having a zero population. A chemical species can catalyze the synthesis of some other chemical species as

$$[i] + [j] \to [i] + 2[j]. \tag{1}$$

with $i, j = 1, \cdots, k$ according to a randomly chosen reaction network (with a connection rate of the catalytic path given by ρ) which is kept fixed throughout each simulation. Furthermore, each molecular species i has a randomly chosen catalytic ability $c_i \in [0, 1]$. (I.e., the above reaction (1) occurs with the rate c_i). Assuming an environment with an ample supply of chemicals available to the cell, the molecules then replicate leading to an increase in their numbers within a cell. It is the dynamics of these molecule numbers N_i of the species i under replication that are our main concern here.

During the replication process, structural changes, e.g. the alternation of a sequence in a polymer, may occur that alter the catalytic activities of the molecules. The rate of such structural changes is given by the replication 'error rate' μ. As a simplest case, we assume that this 'error' leads to all other molecule species with equal probability, (i.e., with the rate $\mu/(k-1)$). In reality of course, even after a structural change, the replicated molecule will keep some similarity with the original molecule, and this equal rate of transition to other molecule species is a drastic simplification. We therefore carried out also some simulations where the errors in replication only lead to a limited range of molecule species and found that the simplification does not affect the basic conclusions presented here.

The model is simulated as follows: At each step, a pair of molecules, say, i and j, is chosen randomly. If there is a reaction path between species i and j, and i (j) catalyzes j (i), one molecule of the species j (i) is added with probability c_i (c_j), respectively. The molecule is then changed to another randomly chosen species with the probability of the replication error rate μ. When the total number of molecules exceeds a given threshold (denoted as N), the cell divides into two such that each daughter cell inherits half ($N/2$) of the molecules of the mother cell, chosen randomly. In order to take the importance of the discreteness in the moluclue numbers [6] into account, we adopted a stochastic rather than the usual differential equations approach.

The cell state at the n-th division is character-

ized by the molecule numbers of the chemical species $\{N_1^n, N_2^n, \cdots, N_k^n\}$ (with $\sum_j N_j^n = N$), while there are four basic parameters; N, k, μ, and ρ. By investigating the dynamics of one thousand randomly chosen networks, and changing the four parameters, we have found that the behaviors of the system can be classified into just the following three types:

(A) Fast switching states without recursiveness

(B) Fixed recursive states

(C) Itinerancy over several quasi-recursive states

In phase (A), even though each generation has some dominating species as with regards to the molecule numbers, the dominating species change every few generations and information regarding the previously dominating species is totally lost often to the point that its population drops to zero. Indeed, by autocatalytic reactions, the population of one dominant species can be amplified, but soon it is replaced by another chemical that is catalyzed by it (see Fig.1a).

In phase (B), a recursive state is established where the chemical composition is stable enough to withstand the division process. Once reached, this state lasts very long (e.g., for as long the simulation lasts, say 10^4 generations) (see Fig.1b).

The recursive state ('attractor') here is not necessarily a fixed point (with fluctuations) since the molecule numbers may oscillate in time. Nevertheless, the overall chemical compositions remain within certain ranges: for example, the major species (i.e. those that are synthesized by themselves, not by an error in the replication process) are not altered over the generations. Generally, all the observed recursive states consist of 5-12 species, except for those species which exist only as a result of replication errors. (see also [10] for recursive transmission of state in a network model with some structure).

For example, in the recursive state depicted in Fig.1b, there are 11 species whose populations remain in existence throughout the simulation. As is shown in Fig.2, the replication of the molecules is sustained by the 'core hypercycle' $109 \to 11 \to 13 \to 109$ where the catalytic activities of these core species satisfy $c_{13} > c_{109} > c_{11}$, and accordingly we have for the respective populations $N_{11} > N_{109} > N_{13}$. This relationship is natural, since molecules with higher catalytic activities result in the synthesis of more molecules other than themselves thus suppressing their own population fraction[8].

Here, through mutual catalyzation, molecules with higher catalytic activities are catalyzed by molecules with lower activities but larger populations. To destroy such a network of mutual support, large fluctuations in molecule numbers are required, which are rare for large N. Hence parasitic molecule species cannot easily invade the core.

In phase (C), the system alternates between quasi-recursive states similar to phase (B) that last for many generations and fast switching states similar to phase (A). The quasi-recursive state itself can be subjected

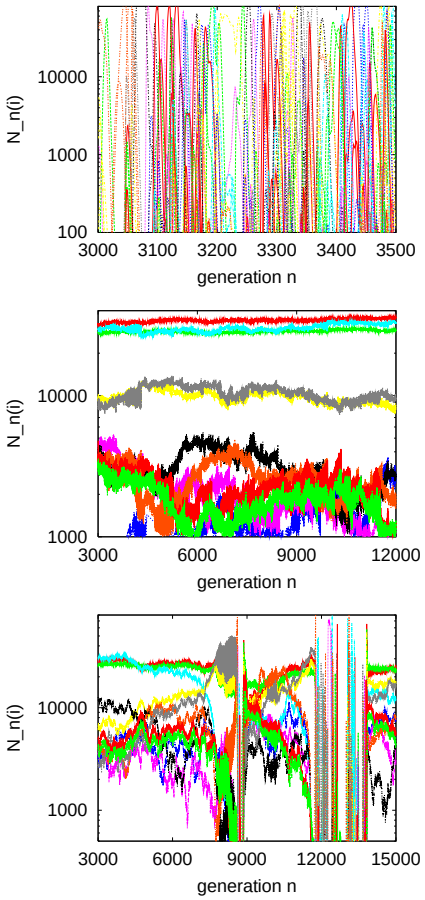


FIG. 1: The number of molecules N_i^n for the species i is plotted as a function of the generation n, i.e., at each successive division event n. $\rho = 0.1$, and $N = 64000$. (a) $\mu = 0.01$, and $k = 500$ (b) $\mu = 0.01$, and $k = 200$ (c) $\mu = 0.1$, and $k = 200$. For (b) and (c), the same network is adopted. Different colors correspond to different species, while only some species (whose population becomes large during some generation) are plotted.

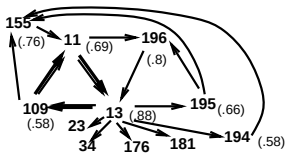


FIG. 2: The catalytic network of the species that constitute the recursive state

to switches between core hypercylces as can be seen in Fig.1c where a switch occurs from an initial core hypercycle (109,11,13), to the next core hypercycle $(11, 13, 195, 155)$ around the 8500th generation. Subsequently, around the 12000th generation, the core network is taken over by parasites to enter the phase (A) like fast switching state which in turn gives way for a new quasi-recursive state around the 14000th generation.

When N is not so large, the molecule number of the

species with the highest catalytic activity in the core hypercycle can become small due to fluctuations, and subsequently succumb to parasitic molecules. Then, the core hypercycle loses its main catalyst resulting in its collapse giving way to a fast switching state that in turn will allow the formation of a new core hypercycle (which can but does not have to be identical to the previous one).

Which one of the phases (A), (B), (C) appears, of course, depends on the parameters and the specific structure of the network. There is however, a clear dependence of the fraction of the networks leading to each of these phases on the parameter values. The fraction of (B) increases and the fraction of (A) decreases for increasing N, or for decreasing k, ρ or μ. For a more systematic investigation, it is useful to classify the phases by the similarity of the chemical compositions between two cell division events[10]. This can be done by defining a k-dimensional vector $\vec{V_n}=(p_n(1),..,p_n(k))$ with $p_n(i) = N_n(i)/N$ and measuring the similarity between ℓ successive generations with the help of the inner product $H_\ell = \vec{V_n} \cdot \vec{V_{n+\ell}} /(|V_n||V_{n+\ell}|)$. In Fig.3, the average similarity $\overline{H_{20}}$ and the average division time are plotted for 50 randomly chosen reaction networks as a function of the path probability ρ. For $\rho > 0.2$, phase (A) is observed for nearly all the networks (e.g. 48/50), while for lower path rates, the fraction of (C) (with a few (B)) increases. (Roughly speaking the networks with $\overline{H_{20}} > .9$ belong to C, and those with $\overline{H_{20}} < .4$ to A).

In general, we have found a positive correlation between the growth speed of a cell, the similarity H, and the diversity of the molecules. (In Fig.3, networks with larger H have smaller division times). The recursive states, established by a variety of species, maintain higher growth speeds since they effectively suppress parasitic molecules. In Fig. 3, for decreasing path rates, the variations in the division speeds of the networks become larger, and some networks that reach recursive states have higher division speeds than networks with larger ρ. On the other hand, when the path rate is too low, the protocells generally cannot grow since the probability to have useful connections in the network is nearly zero. Indeed an optimal path rate seems to exist (e.g., around .05 for $k = 200$, $N = 12800$ as in Fig.3) for which some networks have rather high growth speeds. Consequently, in an environment that necessitates competition for growth, protocells having such optimal networks will be more successful than protocells with sub-optimal networks.

Finally, we investigate the fluctuations of the molecule numbers of each of the species. Since the number of molecules is not very large, the fluctuations over the generations can possibly have a significant impact on the dynamics of the system. In order to quantify the sizes of these fluctuations, have measured the distribution $P(N_i)$ for each molecule species i, by sampling over division events. Our numerical results are summarized as follows:

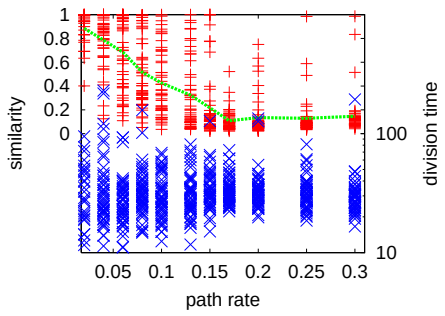


FIG. 3: The average similarity $\overline{H_{20}}$ (red), and the average division time (blue) are plotted as a function of the path rate ρ. For each ρ, data from 50 randomly chosen networks are plotted. The average is taken over 600 division events. The green line indicates the average of $\overline{H_{20}}$ over the 50 networks for each ρ. At $\rho = 0.02$, 25 out of 50 networks cannot support cell growth, 4 cannot at $\rho = 0.04$.

(I) For the fast switching states, the distribution $P(N_i)$ satisfies the power law $P(N_i) \approx N_i^{-\alpha}$, with $\alpha \approx 2$.

(II) For recursive states, the fluctuations in the core network (i.e., 13,11,109 in Fig.2) are typically small. On the other hand, species that are peripheral to but catalyzed by the core hypercycle have log-normal distributions $P(N_i) \approx exp(-\frac{(logN_i-\overline{logN_i})^2}{2\sigma})$, as shown in Fig.4. We have also plotted the variance $\overline{(N_i-\overline{N_i})^2}$ ($\overline{..}$ is the average of the distribution $P(N_i)$) and the deviation between the peak of $P(N_i)$ and $\overline{N_i}$, divided by the average $\overline{N_i}$. As can be seen, the variance and the fluctuations in the core network are small, especially for the minority species (i.e., 13). For molecule species that do not belong to the core hypercycle, the variance scaled by the average increases as the average decreases. Furthermore, there is a distinct deviation between the peak and the average (except for the core species), since the distribution has a tail for larger sizes. On the other hand, if we use the variable $logN_i$ when plotting the distribution, it is closer to a Gaussian, and the difference between the peak and the average is suppressed.

The origin of the log-normal distributions here can be understood by the following rough argument: for a replicating system, the growth of the molecule number N_m of the species m is given by $dN_m/dt = AN_m$ where A is the average effect of all the molecules that catalyze m. We can then obtain the estimate $dlogN_m/dt = \overline{a} + \eta(t)$. by replacing A with its temporal average $\overline{a}$ plus fluctuations $\eta(t)$ around it [9]. If $\eta(t)$ is approximated by a Gaussian noise, the log-normal distribution for $P(N_m)$ is suggested (this argument is valid if $\overline{a} > 0$). For the fast stitching state the growth of each molecule species is close to zero on the average and in this case, by considering the Langevin equation with boundary conditions, the power law follows as discussed in [11, 12].

If several molecules mutually catalyze each other, due

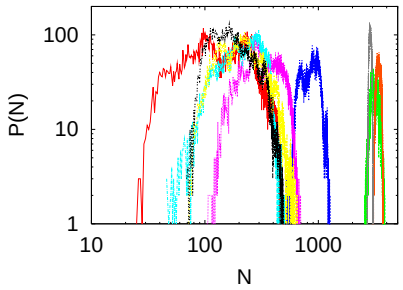


FIG. 4: The number distribution of the molecules corresponding to the network in Fig.2. The distribution is sampled from 1000 division events. From right to left, the plotted species are 11,109,13,155,176,181,195,196,23. Log-Log plot.

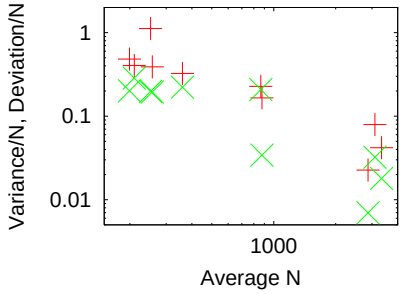


FIG. 5: Scaled variance and deviation. $\times$ denotes the variance of the molecule number divided by its average, while $+$ shows the difference between the average and the peak of the distribution divided by the average $\overline{N_i}$. From the right to left, the species 11,109,13,155,194,176,195,181,196,23,34 are plotted, i.e., the largest $\overline{N_i}$ corresponds to species 11, the third largest to 13, and so forth. Computed in the same way as Fig.4.

to the central limit theorem, one would expect their distributions to be close to Gaussian, and this is indeed the case for the three core species.

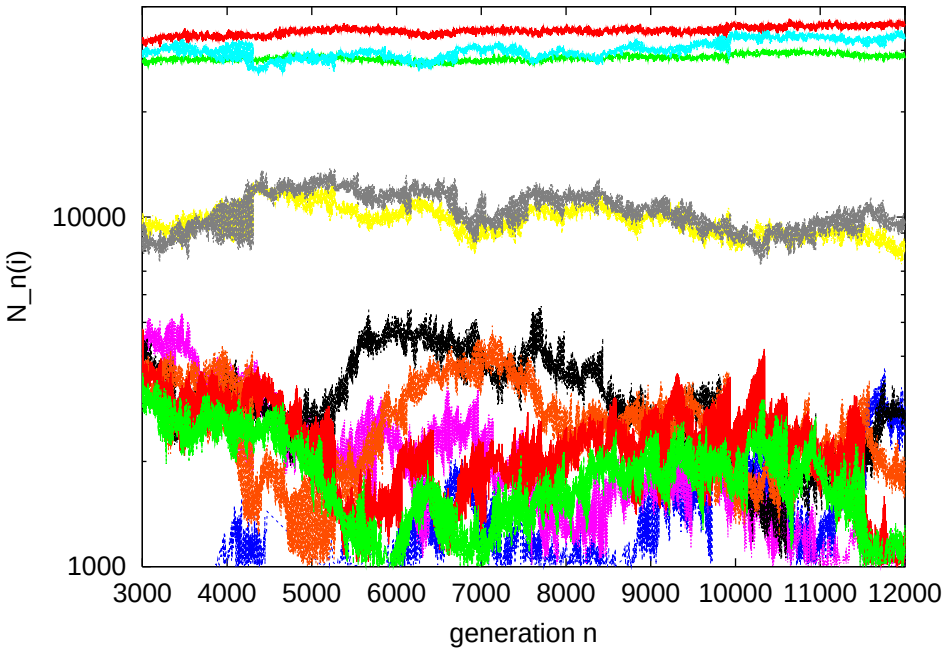
By studying a variety of networks, the observed distributions of the molecule numbers can be generally summarized as: (1)Distribution close to Gaussian form, with relatively small variances in the core (hypercycle) of the network. (2)Distribution close to log-normal, with larger fluctuations for a peripheral parts of the network. (3) Power-law distributions for parasitic molecules that appear intermittently[13].

To sum up, features of a protocell with catalytic reactions and divisions are classified into three phases. Recursive states (B) and switching over quasi-recursive states (C) should be noted, that maintain catalytic activities for cell reproduction. Besides the establishment of recursive growth, variability of cells in their chemical compositions is necessary, in order to ensure evolvability. Previously, we pointed out the relevance of minority molecules in mutually catalyzing systems for making recursive growth and evolution compatible[**?**]. Indeed, phase (C) satisfies both the features, since novel quasi-recursive states with different chemical compositions are visited successively, triggered by extinctions of minority molecules in the core hypercycle networks.

We showed suppression of the fluctuation of molecules at a core hypercycle network and ubiquity of log-normal distribution of those at a peripheral network, which can be testified for the present cell, using recent advances in quantitative measurements of the fluctuations. Indeed, it is interesting to note that the distributions of the abundances of fluorescent proteins, measured by flow-cytometry are often closer to log-normal than Gaussian[14]. Furthermore, e.g. the size of bacteria[14] and some cells in blood[15] (as well as human body weight) obey log-normal distributions.

I would like to thank C. Furusawa, F. Willeboordse, and T. Yomo for useful discussions. This research was supported by 11CE2006.

[1] M. Eigen and P. Schuster, *The Hypercycle* (Springer, 1979).
[2] F. Dyson, *Origins of Life*, Cambridge Univ. Press., 1985
[3] M. Boerlijst and P. Hogeweg, Physica 48D (1991) 17
[4] S.Altmeyer and J.S. McCaskill, Phys. Rev. Lett. 86 (2001) 5819
[5] K. Kaneko, J Biol. Phys., 28 (2002) 781
[6] We use a stochastic approach, since discreteness in molecule numbers is important; Y. Togashi and K. Kaneko, Phys. Rev. Lett., 86 (2001) 2459; see also B. Hess and A. S. Mikhailov, Science **264**, 223 (1994);
[7] K. Kaneko and T. Yomo, J. Theor. Biol. (2002) 312:563
[8] The population sizes of just three species A, B, C, with the catalytic relationship $A \to B \to C \to A$ are estimated by taking the continuum limit $N \to \infty$ and obtaining a fixed point solution of the rate equation for the concentrations of the chemicals. From a straightforward calculation we have: $N_A : N_B : N_C = c_B : c_A : c_C$.
[9] As such this equation diverges with time, but here, the cell divides into two before the divergence becomes significant. The finiteness of the total number N means that the asymptotic distribution as $N \to \infty$ is not available, but the argument on the distribution form is valid as long as N is sufficiently large.
[10] Segre D, Ben-Eli D, Lancet D., Proc. Natl. Acad. Sci. USA 97 (2000)4112
[11] A. S. Mikhailov and V. Calenbuhr, "From Cells to Societies", Springer 2002
[12] D.Sornette, "Critical phenomena in Natural Science", Springer 2002
[13] For some networks, the distributions may sometimes be intermediate between log-normal and Gaussian, and even have double peaks.
[14] T. Yomo, private communication; V. Shivashnakar , private communication
[15] J. Facquet-Danis et al., Biomedicine 25(1970) 294; C.E. Claren et al., Biometrics 47 (1991) 607


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